How much can a fractional audit overstate coalition savings? Exact bounds for pooled newsvendor costs under a standalone-cost floor

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Abstract

Producers that pool their imbalance risk, such as wind farms bidding jointly, share the expected cost of the pooled offer and must check that no group of them would do better on its own. Checking every coalition is hard, so the check is often relaxed to fractional memberships. We ask how far that relaxation can overstate the largest saving available to an actual coalition when every producer is known to pay at least a fraction r of its standalone cost and k producers are excluded from the search. For equal shares we give the exact worst case for every number of producers, every depth k and every r: it equals the elementary ceiling $(1-r)(n-k)/n$ on a union of explicit intervals and is strictly smaller elsewhere, with closed forms in all regimes. For arbitrary shares we give the exact worst case at the deepest nonterminal level and four families of explicit worst-case laws at every depth, whose best value matched a certified upper bound in all 1068 random instances computed. Every bound is attained by a finite demand law. The constructions rest on the fact that costs of symmetric demand laws are zonotope support functions; the upper bounds rest on triangle-inequality certificates. With asymmetric imbalance penalties the ceiling remains valid and every worst case persists; we give the exact worst case at the deepest level for every penalty ratio, and for equal shares show that the windows close once a shortfall costs at least $2(m-1)/m$ times as much as a surplus. For Gaussian demand with nonnegative correlations the overstatement is at most $a_1(1-\sqrt{1-h})$, an explicit bound independent of r (below 0.093 of the pooled cost for up to seven equal producers), and on hourly wind production the observed overstatement is negligible: the bounds act as guarantees rather than predictions.

1 Introduction

Renewable producers that offer their output jointly pay imbalance penalties on the pooled position rather than on their own. Pooling lowers the expected penalty, and the saving must be shared. A cost split is acceptable only if no group of producers would pay less by leaving and offering on its own: the split must lie in the core of the pooling game (Baeyens et al., 2013; Chakraborty et al., 2018). For newsvendor-type games the core is nonempty (Hartman et al., 2000; Müller et al., 2002; Montrucchio et al., 2012), but deciding whether a given split lies in it is NP-hard (Chen and Zhang, 2009), and NP-complete for production–inventory games (Guardiola et al., 2020). Computational work therefore handles the exponentially many coalition constraints by row generation or decomposition (Bonnans and André, 2008; Freire et al., 2015), or approximates the core by sampling its extreme points (Camacho et al., 2025).

A natural tractable check replaces actual coalitions by fractional ones: a producer may join with a fraction of its output, and the largest saving over fractional memberships is a convex program. The fixed-allocation version of this relaxation appears in Burer and Dror (2005, Section 5). Fractional and partially continuous memberships have a long history in cooperative game theory (Aubin, 1981; Samet and Zemel, 1984; Tijs et al., 2004; Shellshear, 2009), and in risk allocation (Denault, 2001; Csóka et al., 2009). Integrality gaps of relaxations are closely tied

to approximate cores in combinatorial games (Deng et al., 1999; Goemans and Skutella, 2004; Caprara and Letchford, 2010). For positively homogeneous costs an efficient allocation passes the full fractional check only if it is a subgradient of the cost extension at the grand coalition, which in the differentiable case is the gradient (Aumann–Shapley) allocation (Denault, 2001; Fischer, 2003). In large newsvendor games with nonatomic demand the core itself shrinks to a single allocation (Montrucchio and Scarsini, 2007). With few producers, however, the fractional check tests membership in a set that can be much smaller than the core. The relaxation can only overstate the saving. The question of this paper is by how much. Worst cases over demand laws also appear in distributionally robust newsvendor games, where they decide whether the core is nonempty (Doan and Nguyen, 2020); here they bound the error of an audit.

Without further information the answer is discouraging: the overstatement can be as large as the charges of the producers kept in the search (Proposition 1). But an auditor usually knows more. We assume that every producer pays at least a fraction r of its standalone cost, and ask for the exact worst case as a function of r, of the shares and of the number k of producers excluded from the search.

Results. For equal shares the answer is complete (Theorems 5 and 6, Figure 1). The elementary ceiling $(1-r)(n-k)/n$ is exact on a union of explicit intervals of r; above $(m-1)/(m+1)$, with $m = n-k$, the worst case is $(m-1)(1-r^2)/(2nr)$; and in the remaining "windows" it is the larger of two explicit families, with a quadratic in r deciding which. For arbitrary shares the worst case at depth $n-2$ has three regimes per pair of producers (Theorem 8). At every depth, four explicit families give lower bounds (Theorem 9) whose maximum matched a certified upper bound in every one of 1068 random instances, which we state as a conjectured closed form (Conjecture 10). With generic shares the elementary ceiling turned out to be exact whenever $r \le 0.4$ in our samples, and never when $r \ge 0.7$, a pattern explained by the criterion of Proposition 17. Section 7 extends the analysis to asymmetric imbalance penalties. Every worst case persists, the last depth is solved for every ratio of the two penalties, and for equal shares the windows close once a shortfall costs at least $2(m-1)/m$ times as much as a surplus; for arbitrary shares, two-block laws give a lower bound that is usually, but not always, exact. On hourly production of three four-producer wind panels the observed overstatement is negligible, at most 0.0013 of the pooled cost, while the worst case for the same shares and floor ranges from 0.005 to 0.41 (Section 9): the bounds are guarantees, useful on their own when r is close to one.

Method. Every lower bound is attained by an explicit finite, centrally symmetric demand law. Such laws have costs $c(x) = \sum_k p_k |x_k^\top x|$, support functions of zonotopes, a classical identity (Koshevoy and Mosler, 1997; Molchanov et al., 2014); the constructions are lists of generators. Every upper bound is a nonnegative combination of triangle-inequality certificates. Requiring positive definite covariance and strict stability of every coalition does not change any of the suprema (Proposition 11).

2 Model

Let $N = \{1, \dots, n\}$, $n \ge 3$, and let $Y = (Y_1, \dots, Y_n)$ be bounded, nonnegative outputs with arbitrary dependence. Imbalance is priced by the symmetric absolute loss $\ell(q,t) = |t-q|$. For a membership vector $z \in \mathbb{R}^n_+$ the *physical fractional cost* is

$$c(z) = \min_{q \in \mathbb{R}} \mathbb{E}|z^\top Y - q|.$$

It is nonnegative, convex and positively homogeneous; with signed arguments it is a seminorm. A frozen grand offer q_0 has expected cost $L = \mathbb{E}|\mathbf{1}^\top Y - q_0| \ge c(\mathbf{1})$, and producer j is charged $a_j = L w_j$ for positive shares w summing to one. We normalize $L = 1$ throughout, so $a = w$.

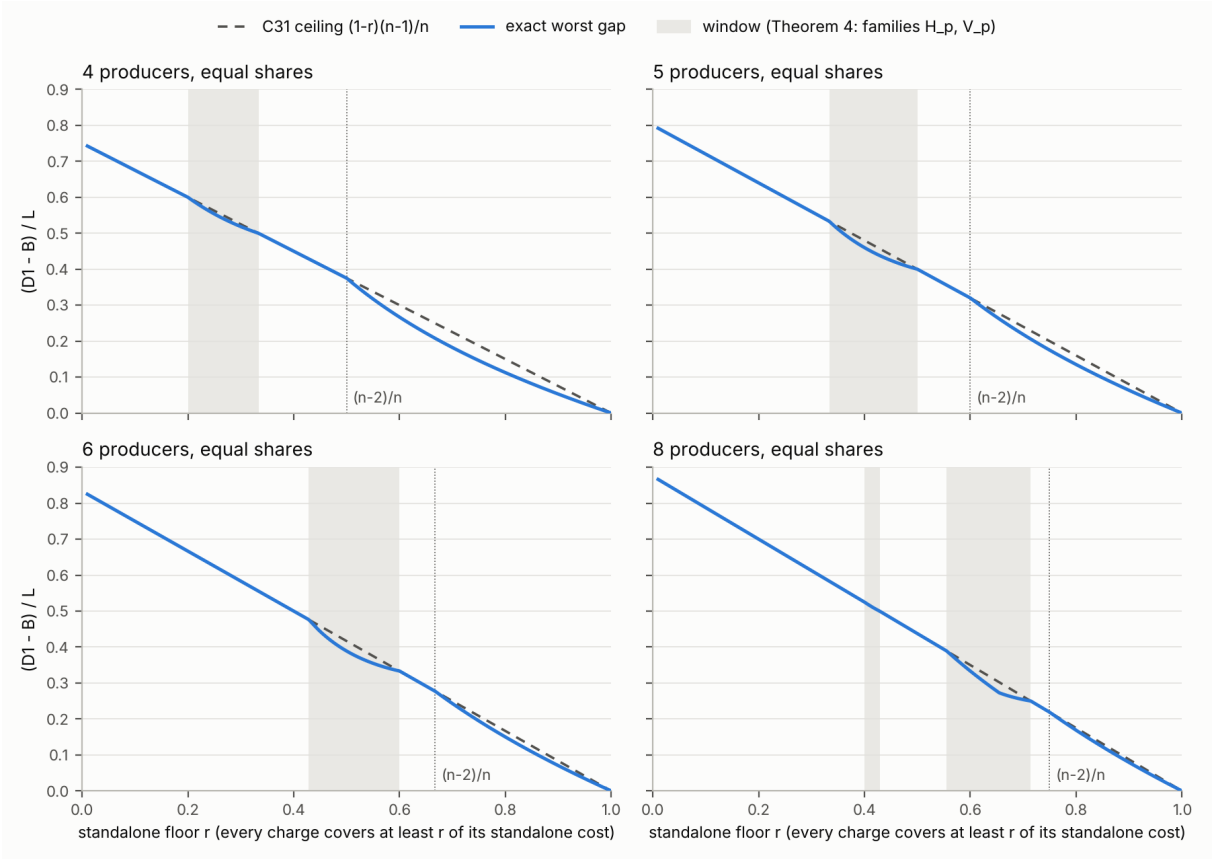


Figure 1: Exact worst-case overstatement $(D_1 - B)/L$ at depth one for equal shares, against the standalone floor r, with the elementary ceiling $(1-r)(n-1)/n$ (dashed). Shaded bands are the windows of Theorem 6; the dotted line marks $(n-2)/n$.

The excess of a membership is $f(z) = a^\top z - c(z)$, the largest positive saving of an actual proper coalition is $B = \max\{0, \max_{\emptyset \neq S \subsetneq N} f(\mathbf{1}_S)\}$, and at depth k the audit computes

$$D_k = \max\Big\{0,\ \max_{|N\setminus S|<k} f(\mathbf{1}_S),\ \max_{|J|=k}\ \max_{z\in[0,1]^n,\ z_J=0} f(z)\Big\},$$

so that $D_k \geq B$ and $D_{n-1} = B$. The *standalone ratio* is $\rho = \min(\{1\} \cup \{a_j/c(e_j) : c(e_j) > 0\})$; the floor $\rho \geq r$ means $c(e_j) \leq a_j/r$ for every j. We study

$$\Gamma_k(w, r) = \sup\{(D_k - B)/L \ :\ \rho \geq r\},$$

the supremum over bounded demand laws and frozen offers. Write $s_k(w)$ for the sum of the $n-k$ largest shares.

Proposition 1 (Elementary ceilings)**.** *$\Gamma_k(w, 0) = s_k(w)$ for $1 \leq k \leq n-2$, and $\Gamma_k(w, r) \leq (1-r)s_k(w)$.*

Proof. On a face with free set F, $f(z) \leq a^\top z \leq a(F) \leq s_k(w)$ because $c \geq 0$; this bound is attained in the limit by two-state laws. For the second statement put $x = \sum_{j\in F} a_j(1 - z_j)$. Nonnegativity gives $f(z) - B \leq a(F) - B - x$; subadditivity and the caps give $c(\mathbf{1}_F) \leq c(z) + x/r$, hence $f(z) - B \leq (1/r - 1)x$. The weighted average of the two bounds with weights $1-r$ and r is $(1-r)(a(F) - B)$. □

The question is how much of the gap between $(1-r)s_k(w)$ and the truth remains, and where. Whatever the answer, it turns a one-sided check into a two-sided one.

Corollary 2 (Two-sided audit)**.** *For every demand law with $\rho \geq r$,*

$$D_k - \Gamma_k(w, r) \;\leq\; B \;\leq\; D_k, \qquad \textit{in particular} \qquad B \;\geq\; D_k - (1-r)s_k(w).$$

Proof. $B \leq D_k$ by construction, and $(D_k - B)/L \leq \Gamma_k(w, r)$ with $L = 1$ by the definition of Γ_k as a supremum over all laws with $\rho \geq r$. □

Both ends are computable: D_k is a convex program on each face, r follows from the standalone costs, and Γ_k is given in closed form by the results below or bounded by the certificate program of Section 3. A fractional audit that reports a saving larger than $\Gamma_k(w, r)$ therefore proves that some actual coalition gains, and by how much at least, without searching the coalitions; an audit that reports a small saving bounds every coalition's gain from above.

3 Two tools

Lemma 3 (Laws from generators)**.** *Let $g_1, \dots, g_m \in \mathbb{R}^n$. The law that picks k uniformly and a sign $\pm$ uniformly and sets $Y = \mu \pm m g_k$ has $c(x) = \sum_k |g_k^\top x|$ for every x. After an affine change of units it takes values in $[0,1]^n$, and its centre is an optimal grand offer with $L = c(\mathbf{1})$.*

Proof. $x^\top Y$ is symmetric about $x^\top \mu$, which is therefore an optimal offer, and $\mathbb{E}|x^\top(Y - \mu)| = \sum_k |g_k^\top x|$. Scaling all outputs, offers and costs by a common factor changes no ratio. □

Hence a value V is attained at depth k as soon as some generators satisfy the caps $c(e_j) \leq a_j/r$, stability $c(\mathbf{1}_S) \geq a(S)$ for every proper S (so $B = 0$), $c(\mathbf{1}) = 1$, and $a^\top z - c(z) = V$ at some $z \in [0,1]^n$ with k zero coordinates. All constructions below are of this form.

Lemma 4 (Triangle certificates)**.** *Let z lie on a face with free set F, let $S \subseteq F$ and $0 \leq \nu \leq 1$. Then*

$$f(z) - B \leq a^\top z - \nu\, a(S) + \sum_{i \in S} \frac{a_i}{r} |z_i - \nu| + \sum_{i \in F \setminus S} \frac{a_i}{r} z_i.$$

Proof. $\nu \mathbf{1}_S = z - (z - \nu \mathbf{1}_S)$, so the triangle inequality and the caps give $\nu c(\mathbf{1}_S) \leq c(z) + \sum_{i \in F} (a_i/r)|z_i - \nu \mathbf{1}_S(i)|$, while $c(\mathbf{1}_S) \geq a(S) - B$. Since $\nu \leq 1$, the term $\nu B - B$ is nonpositive. □

By positive homogeneity a maximizing membership with positive excess can be scaled until its largest coordinate is one. With that coordinate and the order of the others fixed, every certificate with ν equal to a coordinate is linear in z, so the best bound from all of them is a linear program. It is valid for every demand law.

4 Equal shares

Let $a_j = 1/n$, write $M = a/r$ for the common cap, $m = n - k$ for the number of free producers, and for $p = 1, \dots, m-1$ put $q = m - p$ and

$$I_p = \Big[\max\Big(0, \tfrac{q-p}{q+p}\Big),\ \tfrac{q}{q+2p}\Big].$$

Theorem 5. *For equal shares, $1 \leq k \leq n-2$ and $0 < r \leq 1$,*

$$\Gamma_k = \frac{(1-r)m}{n} \quad \textit{if } r \in I_1 \cup \dots \cup I_{m-1}, \qquad \Gamma_k = \frac{(m-1)(1-r^2)}{2nr} \quad \textit{if } r \geq \frac{m-1}{m+1}.$$

Proof. The first value is the ceiling of Proposition 1. For the second, combine for each fractional coordinate the certificates of Lemma 4 with S containing or omitting it ($\nu = 1$): with the largest coordinate at one, $f(z) - B \leq \sum_j h(z_j)$ where $h(t) = \min\{(1-t)(M-a), t(M+a)\}$, and also $f(z) - B \leq a^\top z$. The minimum is concave and symmetric in the $m-1$ other coordinates,

so a common value is optimal, and a one-variable computation gives the stated value for $r \geq (m-1)/(m+1)$.

Construction on I_p. Take p unit producers I, q fractional producers J, the excluded set K, $t = (1-r) - pr/q$, generators ae_l ($l \in K$) and $(Mt/p)\mathbf{1}_I - Me_j$ ($j \in J$), and $z = \mathbf{1}_I + t\mathbf{1}_J$. Every generator vanishes at z, so $c(z) = 0$ and $a^\top z = (1-r)m/n$. The unit caps read $qt \leq p$, i.e. $r \geq (q-p)/(q+p)$; $c(\mathbf{1}) = ka + qM(1-t) = 1$. For a coalition with s unit and τ fractional members (excluded members add a to both terms), $c(\mathbf{1}_S) - a|S| = \tau(M-a) + (q-2\tau)Mts/p - as$, bilinear in (s,τ), with corner values 0, $q(M-a)$, $qMt - ap \geq 0$ (i.e. $r \leq q/(q+2p)$) and 0.

Construction for $r \geq (m-1)/(m+1)$. Take one unit producer u, the others J, $\delta = (M-a)/2$, $\beta = a - (m-1)\delta \geq 0$, generators βe_u, ae_l ($l \in K$), $\delta e_u - Me_j$ ($j \in J$), and $z = e_u + \frac{1-r}{2}\mathbf{1}_J$. Then $c(z) = \beta$, $c(e_u) = a$, $c(e_j) = M$, $c(\mathbf{1}) = 1$, and $c(\mathbf{1}_S) = a|S|$ when $u \in S$ and $c(\mathbf{1}_S) \geq a|S|$ otherwise. The excess at z is the stated value. □

The intervals I_p leave gaps when $p^2 + p < m$: the *window* between I_{p+1} and I_p is (r_L, r_R) with $r_L = (m-p-1)/(m+p+1)$ and $r_R = (m-2p)/m$. At depth one these are $(1/5, 1/3)$ for $n = 4$, $(1/3, 1/2)$ for $n = 5$, $(3/7, 3/5)$ for $n = 6$; their total length is between 0.13 and 0.20 for $4 \leq n \leq 30$ and decays only like $n^{-1/2}$.

Inside a window two further families compete. The family H_p keeps p unit producers and puts the others at $(1-r)/2$, with positive cost at the witness; it is valid for $r \geq (m-p)/(m+p)$ and has value

$$H_p(r) = \frac{(m-p)(1-r^2)}{2nr}.$$

The family V_p moves one producer from unit to fractional: with $q' = m-p-1$,

$$t_1 = \frac{(p+1)(1-r)}{2(p+1) - m(1-r)}, \qquad t_2 = \frac{(p+1)r - 1 + t_1}{q'}, \qquad V_p(r) = \frac{1}{n}\big(p - 1 + (p+1)r + 2t_1\big).$$

Theorem 6. *Let $p^2 + p < m$, $r_L \leq r \leq r_R$ and $Q(r) = mr^2 + 2(p-1)r + 2 + 2p - m$. Then $\Gamma_k = V_p(r)$ if $Q(r) \geq 0$ and $\Gamma_k = H_{p+1}(r)$ if $Q(r) \leq 0$. Equivalently $\Gamma_k = \max(V_p, H_{p+1})$, since $V_p - H_{p+1} = a(m+p+1)(r-r_L)\,Q(r)/(2rD)$ with $D = 2(p+1) - m(1-r) > 0$ on the window.*

Proof. Laws. For H_{p+1} write $p' = p+1$ for the number of unit producers and $q = m - p'$, and use the generators βe_i on the units with $\beta = ((p'+q)a - qM(1+r)/2)/p'$, ae_l on K, and $(\delta/p')\mathbf{1}_I - Me_j$ on the fractional producers, where $\delta = M(1-r)/2$; the witness is $\mathbf{1}_I + \frac{1-r}{2}\mathbf{1}_J$, and the stability slack is again bilinear with corners 0, $q(M-a)$, 0, 0. It is valid because $r \geq r_L = (m-p')/(m+p')$ makes $\beta \geq 0$. For V_p take one producer $*$ in transition and

$$\gamma = \frac{M(p+1)(1-r)}{2p}, \qquad \alpha = \frac{M((p-1) + (p+1)r)}{2q'},$$

$$\beta = \frac{M(m(1-r) - 2p)}{2q'}, \qquad \delta = -\frac{M(2(p+1) - m(1-r))}{2},$$

with generators $(\alpha/p)\mathbf{1}_I + \beta e_* - Me_j$ for the other fractional producers, $\gamma\mathbf{1}_I + \delta e_*$, and ae_l on K, at $z = \mathbf{1}_I + t_1 e_* + t_2\mathbf{1}_{J'}$. All generators vanish at z, every free producer sits exactly at its cap, $c(\mathbf{1}) = 1$, and for each of the two cases $* \in S$ and $* \notin S$ the stability slack is bilinear in the counts, with corner values 0, $q'(M-a)$, $p(M-a)$, $a((m-p)/r - (m+p))$ and $M-a$, $a(p/r+p)$, 0, 0, all nonnegative on the window. The witness is admissible: $t_1 \leq 1$ is $r \geq r_L$, and $t_2 \geq 0$ reduces to a quadratic inequality in $1-r$ that holds on the whole window.

Upper bounds. With Z_I the sum of the p largest coordinates, $u = z_{p+1}$ and Y the sum of the rest, four certificates suffice: nonnegative cost F_0; the $p+1$ largest coordinates with $\nu = 1$ (F_A); the whole face with $\nu = u$ (F_B) and with $\nu = 1$ ($F_{B'}$). If $Q(r) \geq 0$, the weights $\lambda_B = 1$, $\lambda_A = (M(m-2p) - ma)/(2M)$, $\lambda_0 = (M^2(2+2p-m) + 2aM(p-1) + ma^2)/(2aM)$ and the multiplier $M(2+2p-m) + am$ on $p - Z_I \geq 0$ cancel every variable; they are nonnegative because

$r \le r_R$, $Q(r) \ge 0$ and $r \ge r_L$, and the resulting constant is $V_p(r)$. If $Q(r) \le 0$, the weights $\lambda_A = (M-a)/(2M)$ and a split of the remaining $(M+a)/(2M)$ between F_B and $F_{B'}$, with multipliers on $p - Z_I \ge 0$ and $1-u \ge 0$, cancel every variable, and feasibility of nonnegative weights rearranges exactly to $Q(r) \le 0$; the constant is $H_{p+1}(r)$. □

Corollary 7. *For equal shares, $\Gamma_k(w,r)$ is known in closed form for every n, k and r, and is attained by a finite symmetric law with at most n generators. For four producers at depth one,*

$$\Gamma_1 = \tfrac{3}{4}(1-r)\ (r \le \tfrac{1}{5}), \quad \frac{2-r+3r^2}{2(1+3r)}\ (\tfrac{1}{5} \le r \le \tfrac{1}{3}), \quad \tfrac{3}{4}(1-r)\ (\tfrac{1}{3} \le r \le \tfrac{1}{2}), \quad \frac{1-r^2}{4r}\ (r \ge \tfrac{1}{2}).$$

At $r = 1/4$, for instance, four producers who each pay at least a quarter of their standalone cost can still face an audit that reports a saving of $31/56 \approx 55\%$ of the expected pooled cost while no actual coalition gains anything.

5 Arbitrary shares

Theorem 8. *For positive shares and $0 < r \le 1$, $\Gamma_{n-2}(w,r) = \max_{i \ne j} \Phi_{ij}(r)$, where, with $x = a_i/a_j$,*

$$\Phi_{ij}(r) = \begin{cases} a_j(1-r^2)/(2r), & x \ge (1-r)/(2r), \\ (1-r)(a_i+a_j), & (1-r)/(1+r) \le x \le (1-r)/(2r), \\ a_i(1+r)(a_i+a_j)/(ra_j+(1+r)a_i), & x \le (1-r)/(1+r). \end{cases}$$

Proof. At depth $n-2$ a face has two free producers; scale the witness to $z = e_i + te_j$. Nonnegative cost and Lemma 4 with the singleton $\{i\}$ at $\nu = 1$ and the pair $\{i,j\}$ at $\nu = 1$ and at $\nu = t$ bound the excess by $a_i + ta_j$, $ta_j(1+1/r)$, $(1-t)a_j(1/r-1)$ and $(1-t)a_i(1+1/r)$ respectively; the maximum over t of their minimum is Φ_{ij}. The three regimes are attained by the families C31, H and G of Theorem 9 with $|I| = |J| = 1$. □

Fix an excluded set K, a split of the free producers into units I and fractional producers J, and write $a(I)$, A_J for their share totals.

Theorem 9. *For positive shares and any depth k, $\Gamma_k(w,r)$ is at least the largest of the following values over all K and splits:*

(a) C31: $(1-r)(a(I)+A_J)$ if $\frac{1-r}{1+r}A_J \le a(I) \le \frac{1-r}{2r}A_J$;

(b) H: $A_J(1-r^2)/(2r)$ if $a(I) \ge \frac{1-r}{2r}A_J$;

(c) G: $a(I) + tA_J$ with $t = a(I)/(rA_J + (1+r)a(I))$, if $a(I) \le \frac{1-r}{1+r}A_J$.

Proof. In all three families the excluded producers carry generators $a_l e_l$ and the unit coefficients are proportional to the shares. (a) With $t = (1-r) - ra(I)/A_J$, one generator per $j \in J$ with entries $(a_j/r)ta_i/a(I)$ on the units and $-a_j/r$ at j; the witness $\mathbf{1}_I + t\mathbf{1}_J$ has zero cost, the unit caps are the lower condition, and with σ the unit share fraction of a coalition and $a(T)$ its fractional share, the stability slack is bilinear with corners 0, $A_J(1/r-1)$, $A_J t/r - a(I)$ and 0. (b) With $\Delta = A_J(1-r)/(2r)$, unit generators $(1 - \Delta/a(I))a_i e_i$ and fractional generators with entries $(a_j/r)\frac{1-r}{2}a_i/a(I)$ on the units and $-a_j/r$ at j; the slack is $a(T)(1/r-1)(1-\sigma) \ge 0$. (c) With $c_j = a_j a(I)/(rtA_J)$, generators with entries $c_j t a_i/a(I)$ on the units and $-c_j$ at j; the unit caps are tight, $c(\mathbf{1}) = 1$, the witness $\mathbf{1}_I + t\mathbf{1}_J$ has zero cost, and the slack is bilinear with corners 0, $A_J(\lambda-1)$, $a(I)(1/r-1)$, 0, where $\lambda = a(I)/(rtA_J) \ge 1$. □

A fourth family, with one producer in transition as in V_p, has the value $a(I)(1+r) - a_*(1-r) + 2t_1 a_*$ with $t_1 = (1-r)(a(I)+a_*)/(ra(F)+a(I)+a_*-A_{J'})$; we have not yet written general validity conditions for it and check each candidate exactly.

Conjecture 10. *For positive shares, every depth k and $0 < r \le 1$, $\Gamma_k(w, r)$ equals the largest value of the four families over all excluded sets and splits.*

The lower bound is proved, so the conjecture concerns the upper bound. For every instance it can be tested with the certificate linear program of Section 3. In 1000 random instances with up to six producers, random depth and r, the best family was a valid law in exact arithmetic and its value equalled the certificate bound in all 1000 cases (C31 in 482, H in 467, the transitional family in 27, G in 24); the same held in 68 further instances. In those 68, the elementary ceiling $(1 - r)s_k(w)$ was the answer in all 39 instances with $r \le 0.4$ and in none of the 15 with $r \ge 0.7$.

For three free producers much of the conjecture is proved. Fix a face and an ordering $z_u = 1 \ge z_b \ge z_c$. Five combinations of certificates have explicit nonnegative weights: the ceiling, the separable bound for units $\{u\}$, and three bounds built on the block $\{u, b\}$. Together they bound the excess by the value of an attained two-block family on every ordering outside the region $a_u(1 + r) < (1 - r)(a_b + a_c)$, $2r(a_u + a_b) > (1 - r)a_c$, $(1 - r)a_c < 2a_u$. Inside it, the criterion of Proposition 17 with $\kappa = 1$ still settles every face that has a subset with share in $[(1 - r)a(F)/2, (1 - r)a(F)/(1 + r)]$. In a sample of 2400 orderings these arguments prove the conjecture on 2248.

6 Strict stability and positive definite covariance

Proposition 11. *For $0 < r < 1$ the suprema in Theorems 5, 6, 8 and 9 are unchanged if the law must also have positive definite covariance and strictly negative excess for every proper coalition.*

Proof. Let X^s put mass 0.9 evenly on the atoms $\pm s(e_i - e_j)$ over all pairs and mass 0.1 on $\pm\zeta\mathbf{1}$. With $1.8s/n > 1$ every proper coalition has $c^s(\mathbf{1}_S) > a(S)$, and $c^s(\mathbf{1}) \le 1$ for small ζ; its covariance is positive definite. Mix a worst-case law at the ratio $r' = (1-\varepsilon)/\min_j(1/r - \varepsilon c^s(e_j)/a_j)$ with X^s, weights $1 - \varepsilon$ and ε. Costs of centrally symmetric laws mix linearly, so the mixture is strictly stable, respects the caps at r, has grand cost at most one and covariance at least ε times that of X^s, and its excess at the witness tends to the supremum as $\varepsilon \to 0$, by continuity of the closed forms in r. □

7 Asymmetric penalties

In most balancing markets a shortfall is settled at a worse price than a surplus. With penalties λ_+ per unit of surplus and λ_- per unit of shortfall the loss is, up to a constant factor, the pinball loss $\ell_\tau(u) = \tau u^+ + (1 - \tau)u^-$ of level $\tau = \lambda_+/(\lambda_+ + \lambda_-)$, and the coalition cost becomes $c(x) = \min_q \mathbb{E}\,\ell_\tau(x^\top Y - q)$, a scaled CVaR-type deviation of $x^\top Y$ in the sense of Rockafellar et al. (2006). Single producers facing such penalties are studied by Pinson et al. (2007) and Bitar et al. (2012); with linear penalties the optimal offer is a quantile, by the newsvendor argument. Write $\kappa = \max(\tau, 1 - \tau)/\min(\tau, 1 - \tau) \ge 1$ for the ratio of the two penalties and $\Gamma_k^\kappa(w, r)$ for the worst gap; $\kappa = 1$ is the symmetric case.

Lemma 12. *The cost c is sublinear and nonnegative, and $c(-x) \le \kappa\, c(x)$ for every x.*

Proof. $\ell_\tau(-u) = \ell_{1-\tau}(u) \le \kappa\, \ell_\tau(u)$ pointwise. □

Call (d, α, β) with $\alpha, \beta \ge 0$ and $\alpha/\beta \in [1/\kappa, \kappa]$ a κ-segment, with cost $\alpha(d^\top x)^+ + \beta(d^\top x)^-$.

Proposition 13. *Every finite sum of κ-segment costs is, up to a positive factor, the pinball cost of a finite demand law on $[0, 1]^n$.*

Proof. Let $\tau \le 1/2$. For a segment, solve $\alpha = \tau A + (1 - \tau)B$, $\beta = (1 - \tau)A + \tau B$; then $A, B \ge 0$ exactly when $\alpha/\beta \in [1/\kappa, \kappa]$. A variable V equal to A/τ and $-B/\tau$ with probability τ each and

to 0 otherwise has 0 as a τ- and a $(1-\tau)$-quantile, and $\mathbb{E}\,\ell_\tau(sV) = \alpha s^+ + \beta s^-$ for every real s. Mixing such components along the directions d keeps 0 optimal for each of them, so their costs add; an affine map puts the law in the box. □

Corollary 14. *For every $\kappa \geq 1$: (a) $\Gamma_k^\kappa(w,r) \leq (1-r)s_k(w)$; (b) every explicit law of Sections 4–6 has a counterpart with the same normalized costs, so every lower bound proved for $\kappa = 1$ holds for every κ.*

Proof. (a) The proof of the ceiling uses only subadditivity along the nonnegative difference $\mathbf{1}_F - z$. (b) Symmetric generators are κ-segments with $\alpha = \beta$. □

In the certificates of Lemma 4 a coordinate that has to be lowered is a shortfall direction, so it is charged $\kappa a_i/r$ instead of a_i/r per unit. With this one change the last-depth and equal-share arguments go through.

Theorem 15. *For positive shares, $0 < r \leq 1$ and $\kappa \geq 1$, $\Gamma_{n-2}^\kappa(w,r) = \max_{i\neq j} \Psi_{ij}(r)$, where, with $x = a_i/a_j$,*

$$\Psi_{ij}(r) = \begin{cases} a_j(1-r)(r+\kappa)/(r(1+\kappa)), & x \geq \kappa(1-r)/(r(1+\kappa)), \\ (1-r)(a_i+a_j), & (1-r)/(r+\kappa) \leq x \leq \kappa(1-r)/(r(1+\kappa)), \\ a_i(r+\kappa)(a_i+a_j)/(a_i(r+\kappa)+ra_j), & x \leq (1-r)/(r+\kappa). \end{cases}$$

For $\kappa = 1$ this is Theorem 8.

Proof. With $z = e_i + te_j$, the certificates for the pair at $\nu = 1$ and $\nu = t$, for $\{i\}$ at $\nu = 1$, and nonnegative cost bound the excess by $(1-t)a_j(1-r)/r$, $(1-t)a_i(r+\kappa)/r$, $ta_j(r+\kappa)/r$ and $a_i + ta_j$; the maximum over t of their minimum is Ψ_{ij}. The other producers get the segments (e_k, a_k, a_k), and one or two segments on the pair attain the three regimes. In the first, the segment along $te_i - e_j$ has $(\alpha,\beta) = (\kappa a_j/r, a_j/r)$ and $t = (1-r)/(1+\kappa)$, and a symmetric segment of weight $a_i - \kappa t a_j/r$ lies along e_i. In the second, the segment along $te_i - e_j$ has $\beta = a_j/r$, $t = 1 - r - rx$ and α in an interval that is nonempty exactly in that regime. In the third, the segment along $e_i - e_j/t$ has $(\alpha,\beta) = (a_i/r, \kappa a_i/r)$. □

For equal shares put $m = n-k$, $H_p^\kappa(r) = (m-p)(1-r)(r+\kappa)/(nr(1+\kappa))$ and

$$I_p^\kappa = \Big[\max\Big(0, \tfrac{m-(1+\kappa)p}{m}\Big),\ \tfrac{\kappa(m-p)}{\kappa m+p}\Big],$$

which for $\kappa = 1$ are H_p and I_p.

Theorem 16. *For equal shares and $\kappa \geq 1$, $\Gamma_k^\kappa(r) \leq \min\{(1-r)m/n,\ H_1^\kappa(r)\}$ for every r, with equality on every I_p^κ and for $r \geq \kappa(m-1)/(\kappa m+1)$. The gap between I_{p+1}^κ and I_p^κ is nonempty if and only if $\kappa < (p+1)(m-p)/(pm)$. In particular, if $\kappa \geq 2(m-1)/m$, and so whenever a shortfall costs at least twice as much as a surplus,*

$$\Gamma_k^\kappa(r) = \min\Big\{(1-r)\frac{m}{n},\ \frac{(m-1)(1-r)(r+\kappa)}{nr(1+\kappa)}\Big\} \qquad (0 < r \leq 1).$$

Proof. With $1 = z_1 \geq z_2 \geq \dots$ and $\sigma = z_2 + \dots + z_m$, the certificates for the face at $\nu = 1$ and for $\{1\}$ at $\nu = 1$ bound the excess by $(1-r)(m-1-\sigma)/(nr)$ and $\sigma(r+\kappa)/(nr)$; their minimum is at most H_1^κ, with equality at $\sigma = (1-r)(m-1)/(1+\kappa)$. The ceiling law of Theorem 5, with p units and the others at $t = 1 - mr/(m-p)$, becomes one segment along $(t/p)\mathbf{1}_I - e_j$ per fractional producer with $\beta = 1/(nr)$ and α in an interval that is nonempty exactly on I_p^κ. The law H_p becomes the same directions with $(\alpha,\beta) = (\kappa,1)/(nr)$ and $t = (1-r)/(1+\kappa)$, plus a symmetric segment on each unit. In both, the stability slack is bilinear in the numbers of units and of fractional producers in a coalition, so the corners suffice. The window condition follows from $(m-(1+\kappa)p)(\kappa m+p+1) - m\kappa(m-p-1) = (1+\kappa)\big(m(p+1-p\kappa) - p(p+1)\big)$. □

For arbitrary shares at intermediate depths the same segments, applied to two blocks of producers, give a lower bound. For an excluded set K and a split of $F = N \setminus K$ into units I and fractional producers J, give each $j \in J$ one segment along $(ta_i/a(I))_{i\in I} - e_j$ with coefficients proportional to a_j, as in the three regimes of Theorem 15. The cost of a coalition then depends only on $a(S\cap I)/a(I)$ and $a(S\cap J)$, and its stability slack is bilinear in these two numbers. Hence

$$\Gamma^{\kappa}_{k}(w,r) \;\geq\; \max_{K,\,I}\; \Psi\big(a(I),\, a(F\setminus I)\big), \tag{1}$$

with Ψ the function of Theorem 15 evaluated at the two share totals; for $\kappa = 1$ this is the best of the families C31, H and G of Theorem 9. In 600 random instances at intermediate depths ($n \leq 6$), the bound (1) equals the certificate bound in all 299 instances with $\kappa \geq 2$, but equality does not hold in general. For shares $(1, 24, 24, 1)/50$, depth one, $r = 33/50$ and $\kappa = 10$, the best value of (1) is 0.2496, while an explicit law with three levels, verified in exact arithmetic, reaches 0.2664; the certificate bound is 0.2787. Transitional laws therefore matter for arbitrary shares at every κ, although they are rare in random samples. What remains true for every $\kappa \geq 1$ is a criterion for the ceiling.

Proposition 17. *Suppose a largest face F contains a nonempty proper subset I with $(1 - r)a(F)/(1 + \kappa) \leq a(I) \leq \kappa(1 - r)a(F)/(r + \kappa)$. Then $\Gamma^{\kappa}_{k}(w, r) = (1 - r)s_k(w)$. This holds in particular when every producer of F has share at most $(1 - r)(\kappa^2 - r)a(F)/((r + \kappa)(1 + \kappa))$.*

Proof. The two ends of the interval are the ends of the ceiling regime of Ψ at the split $(I, F \setminus I)$, so (1) gives the lower bound, and Corollary 14(a) gives the upper bound. Add the producers of F one at a time. If every share is at most the width of the interval, the first partial sum that reaches the interval cannot jump over it, and the interval ends below $a(F)$. □

For $\kappa = 1$ the interval is $[(1 - r)/2, (1 - r)/(1 + r)]\, a(F)$. It shrinks from $[0.4, 0.67]\, a(F)$ at $r = 0.2$ to $[0.15, 0.18]\, a(F)$ at $r = 0.7$, in line with the pattern reported after Conjecture 10.

For four producers at depth one and $r = 1/4$ the worst gap rises from $31/56$ with symmetric penalties to $9/16$ with $\kappa = 2$. Asymmetry never helps: every exact value above is nondecreasing in κ, and as $\kappa \to \infty$ the elementary ceiling becomes exact on $[0, (m - 1)/m]$ for equal shares. Inside the gaps that remain for $1 < \kappa < 2(m - 1)/m$, the certificate bound equals the relaxation over all sublinear costs with $c(-x) \leq \kappa c(x)$. The best κ-segment law we found stays up to $1.3 \cdot 10^{-3}$ below it, so these narrow windows remain open.

8 Gaussian demand

The worst cases above are attained by atomic, carefully correlated laws. For Gaussian demand $Y \sim N(\mu, \Sigma)$ the optimal expected pinball loss of a membership is $\phi(\Phi^{-1}(\tau))\,(x^{\top}\Sigma x)^{1/2}$, a Euclidean norm, symmetric in x whatever the penalty ratio. Write $c(x) = (x^{\top}Gx)^{1/2}$ for a scaled covariance G. The caps, the stability of every proper coalition and the grand cost are then linear constraints on G, so for a fixed membership the least favourable covariance solves a semidefinite program.

Theorem 18. *Let demand be Gaussian with nonnegative correlations and the allocation be in the core. For a membership z on a face, order the face so that $1 = z_1 \geq z_2 \geq \cdots \geq z_m \geq 0$, and put $A_i = a_1 + \cdots + a_i$ and $\gamma_i = A_i^2 - A_{i-1}^2$. Then*

$$a^{\top}z - c(z) \;\leq\; \sum_i a_i z_i - \Big(\sum_i \gamma_i z_i^2\Big)^{1/2} \;\leq\; a_1\big(1 - \sqrt{1 - h}\big), \qquad h = \sum_{i\geq 2} \frac{a_i}{a_i + 2A_{i-1}},$$

the second inequality when $h < 1$, whatever the floor r. The bound is attained by independent demand with variances γ_i whenever $r \leq \min_i \sqrt{a_i/(a_i + 2A_{i-1})}$ and the stationary membership $z_i = a_1 a_i/(\gamma_i\sqrt{1 - h})$ lies in the cube.

panel	allocation	r	B/L	depth one		depth two	
				observed	worst case	observed	worst case
Texas	proportional	0.97–0.98	0.0005	0.0013	0.012–0.025	0.0001	0.006–0.013
Texas	gradient	0.95–0.98	0.0000	0.0000	0.017–0.038	0.0000	0.009–0.020
France	proportional	0.98–0.99	0.0011	0.0002	0.005–0.008	0.0000	0.003–0.004
France	gradient	0.98–0.99	0.0000	0.0000	0.007–0.012	0.0000	0.004–0.006
Four states	proportional	0.51–0.55	0.0000	0.0000	0.349–0.371	0.0000	0.183–0.192
Four states	gradient	0.47–0.51	0.0000	0.0000	0.383–0.409	0.0000	0.200–0.223

Table 1: Observed overstatement $(D_k - B)/L$ of the fractional audit on wind data (largest value over two years and $\kappa \in \{1, 2\}$) and the worst case for the same shares and floor r (range; depth one: best explicit family to certificate bound; depth two: exact, Theorems 8 and 15).

Proof. Write $z = \sum_s \mu_s \mathbf{1}_{U_s}$ with U_s the s largest coordinates and $\mu_s = z_s - z_{s+1} \geq 0$. For $s \leq t$, nonnegative correlations give $\mathbf{1}_{U_s}^\top G \mathbf{1}_{U_t} \geq \mathbf{1}_{U_s}^\top G \mathbf{1}_{U_s}$, and stability of U_s gives $\mathbf{1}_{U_s}^\top G \mathbf{1}_{U_s} \geq A_s^2$. Hence $z^\top G z \geq \sum_{s,t} \mu_s \mu_t A^2_{\min(s,t)} = \sum_i \gamma_i z_i^2$. The middle expression is concave in $(z_2, \ldots, z_m)$ and its stationary point gives the last value. Independent demand with variances γ_i, with the excluded producers appended to the order, makes every proper coalition stable, since $\sum_{i \in T} \gamma_i \geq a(T)^2$ for every T, and has $c(\mathbf{1}) = 1$. Its caps are the condition on r, and the excess at the stationary membership equals the bound. □

For equal shares $\gamma_i = (2i-1)a^2$ and the bound is $a(1-\sqrt{1 - H_m})$ with $H_m = \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{2m-1}$: at depth one 0.061, 0.079, 0.086, 0.090 and 0.093 of the pooled cost for $n = 3, \ldots, 7$, attained for $r \leq 1/\sqrt{2n-1}$. Over all laws the corresponding worst cases tend to $(n-1)/n$ as $r \to 0$. Without the sign restriction Gaussian laws can be as bad as arbitrary ones when r is small (for $n = 4$ at depth one and $r = 0.1$ both give 0.675), but they fall far below them as r grows (0.116 against 0.375 at $r = 0.5$).

9 An illustration on wind data

The bounds above are worst cases. To see what an audit finds on actual production, we took three panels of four producers: four nearby sites of the NREL WIND Toolkit in Texas (modeled hourly production, 2007 and 2008, capacities 2, 16, 10 and 16 MW), the four 2.05 MW turbines of the ENGIE La Haute Borne farm in France (measured, 2014 and 2015), and four distant 16 MW WIND Toolkit sites in Texas, Iowa, Wyoming and New York (modeled, 2007 and 2008). For these shares and floors the worst case over Gaussian laws with nonnegative correlations (Section 8) is 0.004 to 0.0125 for the Texas panel, 0.0008 to 0.0018 for the French one and 0.075 to 0.080 for the distant sites, against 0.35 to 0.41 over all laws. For each calendar year the empirical joint law of hourly production defines the costs, with symmetric penalties ($\kappa = 1$) and with a shortfall penalty twice the surplus penalty ($\kappa = 2$), and $L = c(\mathbf{1})$. Two allocations are audited: charges proportional to standalone costs, which give every producer the same ratio $r = c(\mathbf{1})/\sum_j c(e_j)$, and the gradient allocation $a_j \propto \mathbb{E}[Y_j \, \ell'(S - q^*)]$, with S the pooled production and q^* its optimal offer. The excesses B and D_k are computed exactly, by one linear program per face over all hours. Table 1 compares the observed overstatement $(D_k - B)/L$ with the worst case for the same shares and the same floor r.

Three observations stand out. Within the first two panels the producers are close to each other and strongly correlated: pooling saves only 1 to 3.5 percent of the standalone costs, the floor r lies between 0.95 and 0.99, and the worst case is correspondingly small, at most 0.038 of the pooled cost at depth one. A floor this high is therefore a certificate on its own: whatever the demand law, the fractional audit cannot err by more than a few percent. The four distant sites pool far more (the pooled cost is about half the sum of the standalone costs), and the worst case

for the same shares and floor rises to between 0.35 and 0.41 at depth one. Finally, on all three panels the observed overstatement is negligible: at most 0.0013 of the pooled cost, zero for the gradient allocation, as the introduction predicts, and zero for both allocations on the distant sites. By Corollary 2 the fractional audits alone certify that no coalition of these producers gains more than $D_1 \leq 0.0018$ of the pooled cost, without enumerating coalitions. Actual production is far from the extremal laws of the theorems. The bounds are guarantees, not predictions. When r is close to one they certify the fractional audit without any knowledge of the joint law. When r is low, as for a diverse portfolio, they show that no certificate can be obtained from r alone, and the audit has to be checked against the joint law, as was done here.

10 Discussion

The elementary ceiling $(1 - r)s_k(w)$ is the right answer more often than one might expect: for equal shares on most of the range of r, and for generic shares whenever producers pay a small or moderate part of their standalone cost. Where it is not, the exact curves show how much an auditor may discount a fractional audit, and the windows show that the correction is not confined to large r when shares are symmetric. Even with a substantial standalone floor the audit can report savings of half the pooled cost while no coalition gains.

These worst cases are attained by atomic, carefully correlated laws. On hourly wind production (Section 9) the audit was essentially exact, even where the worst case for the same floor exceeded a third of the pooled cost. The results are therefore best read as guarantees. A floor close to one certifies a fractional audit whatever the joint law; a low floor does not, and the audit must then be checked against the joint law, which is cheap once that law is available. Asymmetric penalties make the worst cases larger but do not change this picture.

Three questions remain open. The upper bound in Conjecture 10; whether the relaxation over all seminorms, which the explicit laws met in every case we computed, always coincides with the worst case over demand laws; and the narrow windows that asymmetric penalties leave open in Section 7, together with the exact worst case for arbitrary shares under asymmetric penalties, where transitional laws matter at every penalty ratio. Pinball costs are support functions of reflected sections of lift zonoids (zonoid trimmed regions, Koshevoy and Mosler, 1997), and inequalities satisfied by all such sections would sharpen the relaxation there.

Reproducibility. Every construction, closed form and computational claim is checked by scripts that write their outcomes to JSON before any summary is generated; exact rational arithmetic is used for all constructions, and the algebra of every proof (identities, corner values, dual weights and the sign factorizations) is verified symbolically for general sizes. The wind data of Section 9 are public and verified by checksums before use.

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